

packages/graph-core/src/analyzer/language-lesson.ts

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Overview

A new file `packages/graph-core/src/analyzer/language-lesson.ts` is added, exposing a public API for generating language-specific lesson prompts, detecting concepts, and parsing LLM responses into a typed result.

Key changes

- **Imports:** `GraphNode`, `GraphEdge` from `../types.js` (R1) and `LanguageConfig` from `../languages/types.js` (R2).
- **LanguageLessonResult interface** (lines 4-7) defines `languageNotes: string` and `concepts: Array<{name: string; explanation: string}>` (R4-R7).
- **Base concept patterns** (`BASE_CONCEPT_PATTERNS`, lines 13-42) and `buildConceptPatterns` (lines 48-63) merge language-agnostic and language-specific keywords.
- **detectLanguageConcepts** (lines 69-92) scans node tags, summary, and `languageNotes` for pattern matches.
- **getLanguageDisplayName** (lines 99-107) resolves a human-readable name from `LanguageConfig` or capitalizes the language string.
- **buildLanguageLessonPrompt** (lines 112-147) constructs an LLM prompt that includes node metadata, relationships, and detected concepts.
- **extractJson** (lines 160-172) extracts a JSON block from an LLM response, handling markdown fences.
- **parseLanguageLessonResponse** (lines 178-210) parses the response into `LanguageLessonResult`, returning defaults on failure.

Impact

- Provides a typed contract for lesson data, reducing downstream type errors.
- Centralizes concept detection logic; new language configs can be added without modifying prompt logic.
- Prompt construction is linear in node size; overhead is negligible for typical graph nodes (unknown from the available diff/scan evidence).

Risks & follow-ups

- `parseLanguageLessonResponse` silently defaults on parse errors; add unit tests to verify graceful handling of malformed JSON.
- Accuracy of `detectLanguageConcepts` depends on correct merging of base and language-specific patterns; verify overlap handling.

- Prompt length may grow with many relationships; consider trimming or summarizing relationships if token limits are a concern.